Project Proposal • September 28, 2026

Submitted by: Dimitri Chrysafis **Course:** COMPSCI 566

Monocular Depth Estimation: Benchmarking and a Defocus-Informed Refinement

Team Members

Dimitri Chrysafis
NetID: chrysafis

Vincent Zhou
NetID: vzhou2

The Problem

Take a photo of a chair. Nothing in the pixels says whether it is a real chair two meters away or a toy chair on a desk, because shrinking a scene and moving it closer leaves the image almost unchanged. That scale ambiguity is the core difficulty of monocular depth estimation. We want a dense depth map from one RGB image and, given a short focal stack, depth in real meters.

Modern networks sidestep scale. They predict relative depth, known only up to an affine map of inverse depth, $z = 1/(a\,d + b)$, and treat blur as noise to be robust against.

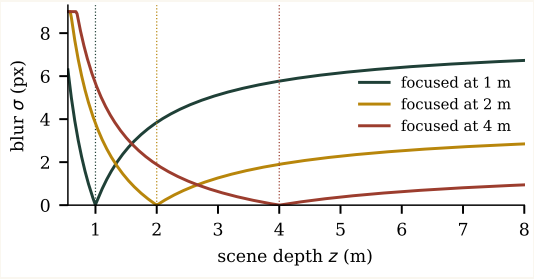


Fig. 1. Thin-lens blur vs. depth for a 25 mm f/2 lens. Blur is zero at the focus plane and grows on both sides, so a few focus settings pin down depth in meters.

Blur is actually the one monocular cue with absolute scale. Under the thin-lens model, a point at depth z spreads into a Gaussian with standard deviation

$$\sigma(z) = k\left|1 - \frac{z_f}{z}\right|, \qquad k = \frac{f_{\text{px}}\, f}{2N\,(z_f - f)},$$

where f_{px} is the focal length in pixels, f the physical focal length, N the f-number, and z_f the focus distance. Everything on the right is a lens constant, so measured blur maps straight to meters (Fig. 1).

We care about this because depth is what turns a phone camera into a 3D sensor (portrait mode, AR, robot navigation), and because the project joins the two halves of this course: the optics of image formation and focus, and modern learned vision. It also comes with standard metrics, so gains are measurable rather than anecdotal.

State of the Art

Relative monocular depth is close to solved. MiDaS [1] and DPT [2] trained on mixed datasets with a scale-invariant loss; Depth Anything [3] scaled that recipe to 62M unlabeled images; Depth Anything V2 [4] trains on synthetic labels and ships 25M to 1.3B parameter models that run over 10× faster than the diffusion-based Marigold [5]. ZoeDepth [6] bolts on metric heads trained per domain, and Monodepth2 [7] is the standard self-supervised baseline. None of these uses defocus.

Depth from defocus (DFD) goes the other way. Pentland [8] and Subbarao and Surya [9] invert the thin-lens model from local blur, and DDFF [10] and Maximov et al. [11] learn the

blur-to-depth mapping from focal stacks. Stereo methods such as SGM [12] use a second view instead.

Family	Output	Where it fails
Learned monocular [1–7]	relative depth	metric scale; ZoeDepth [6] only within its training domain
Depth from defocus [8–11]	metric depth	textureless regions; needs a stack and calibration
Stereo matching [12]	metric depth	needs two calibrated views; textureless regions

The first two rows fail in complementary places, and we have not found work that fuses a frozen zero-shot model with thin-lens blur at test time. That is our gap.

Approach: Re-implement, Then Improve

Phase 1 (re-implementation). Run Depth Anything V2 (ViT-S) [4] zero-shot on NYU-Depth v2 and DDFF-12, scored after least-squares scale-and-shift alignment as in [1]. Write classical depth from focus from scratch (modified Laplacian focus measure, per-pixel peak with sub-frame interpolation), scored on DDFF-12. SGM [12] is a stretch goal.
Phase 2 (new). Estimate per-pixel blur $\hat{\sigma}$ from the stack, then solve for metric depth as an affine rescale of the network's relative depth $\hat{d}$ plus a smooth residual r:

$$z(\mathbf{x}) = \frac{1}{s\,\hat{d}(\mathbf{x}) + t + r(\mathbf{x})}, \qquad \min_{s,t,r} \sum_{\mathbf{x}} w(\mathbf{x})\,\rho(\sigma(z(\mathbf{x})) - \hat{\sigma}(\mathbf{x})) + \lambda\|\nabla r\|^2.$$

Here ρ is a Huber loss and w is the local focus-measure contrast, so pixels with no texture carry no weight and inherit shape from the network. The global s, t fix scale; r fixes local shape errors a single affine map cannot. It is a small per-image optimization with no retraining and no labels. This should beat both parents because the parameterization matches what each source knows. The network's output is exactly an affine-invariant inverse depth, which is the family blur measurements can pin to metric scale, and the weight w keeps unreliable blur out.

Preliminary Result

We already have a working depth-from-focus baseline. To test it before touching real data, we rendered a 10-frame focal stack (focus from 0.9 to 7 m) of a synthetic scene: a tilted city backdrop from 1.8 to 6 m, a painting at 1.1 m, and a plain grey card at 3 m, blurred layer by layer with the thin-lens model above.

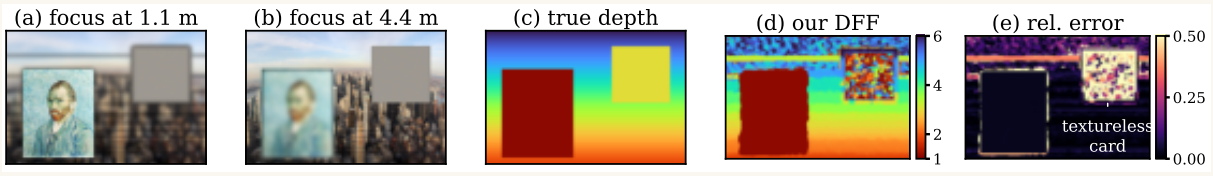


Fig. 2. Our depth-from-focus baseline on a rendered 10-frame stack. (a, b) two frames of the stack; (c) true depth; (d) our estimate; (e) relative error. Textures come from the HW1 images.

On textured pixels, AbsRel is 0.085 with $\delta_{1.25} = 0.90$. On the textureless card it jumps to 0.52, since there is nothing for a focus measure to lock onto. That failure is precisely the region where

a learned prior is reliable, which is the case for Phase 2.

Evaluation and Plan

Data. The 654-image NYU-Depth v2 test split [13]; DDFF 12-Scene [10] (real 10-frame Lytro focal stacks with depth ground truth; we evaluate on its labeled scenes); rendered stacks like Fig. 2 for controlled tests; our own phone focal stacks for qualitative results.
Metrics. AbsRel, SqRel, RMSE, RMSE-log, $\delta < 1.25, 1.25^2, 1.25^3$, plus metric scale error mean $|\log(z_{\text{pred}}/z_{\text{gt}})|$, reported separately on textured and textureless pixels.
Comparisons. DA-V2 raw and oracle-aligned (an upper bound on what scale fitting can do), our DFF, and the fusion. Ablations remove the residual r, the weight w, and the smoothness term, and vary the frame count (2, 3, 5, 10). We also report runtime.

When	To do
Sep 29	Submit proposal.
Sep 30–Oct 9	Set up the repo; DA-V2 on NYUv2 test; alignment and metrics script; first numbers.
Oct 10–20	Port our DFF to DDFF-12; error analysis on both baselines.
Oct 21–29	Freeze Phase 1 numbers; write the midterm report (due Oct 29).
Oct 30–Nov 13	Implement the fusion; ablations on r, w, λ, and frame count.
Nov 14–24	Capture our own phone focal stacks; final runs; build the project webpage.
Nov 25–Dec 10	Presentation (Dec 1–10); final webpage and code (Dec 10).

If the fusion does not help, the Phase 1 benchmark and the textured vs. textureless error analysis still stand as results.

References

[1] R. Ranftl, K. Lasinger, D. Hafner, K. Schindler, V. Koltun. Towards Robust Monocular Depth Estimation: Mixing Datasets for Zero-Shot Cross-Dataset Transfer. *TPAMI*, 2022.

[2] R. Ranftl, A. Bochkovskiy, V. Koltun. Vision Transformers for Dense Prediction. *ICCV*, 2021.

[3] L. Yang, B. Kang, Z. Huang, X. Xu, J. Feng, H. Zhao. Depth Anything: Unleashing the Power of Large-Scale Unlabeled Data. *CVPR*, 2024.

[4] L. Yang, B. Kang, Z. Huang, Z. Zhao, X. Xu, J. Feng, H. Zhao. Depth Anything V2. *NeurIPS*, 2024.

[5] B. Ke, A. Obukhov, S. Huang, N. Metzger, R. C. Daudt, K. Schindler. Repurposing Diffusion-Based Image Generators for Monocular Depth Estimation. *CVPR*, 2024.

[6] S. F. Bhat, R. Birkl, D. Wofk, P. Wonka, M. Müller. ZoeDepth: Zero-shot Transfer by Combining Relative and Metric Depth. *arXiv*, 2023.

[7] C. Godard, O. Mac Aodha, M. Firman, G. J. Brostow. Digging Into Self-Supervised Monocular Depth Estimation. *ICCV*, 2019.

[8] A. P. Pentland. A New Sense for Depth of Field. *TPAMI*, 1987.

[9] M. Subbarao, G. Surya. Depth from Defocus: A Spatial Domain Approach. *IJCV*, 1994.

[10] C. Hazirbas, S. Soyer, M. C. Staab, L. Leal-Taixé, D. Cremers. Deep Depth From Focus. *ACCV*, 2018.

[11] M. Maximov, K. Galim, L. Leal-Taixé. Focus on Defocus: Bridging the Synthetic to Real Domain Gap for Depth Estimation. *CVPR*, 2020.

[12] H. Hirschmüller. Stereo Processing by Semiglobal Matching and Mutual Information. *TPAMI*, 2008.

[13] N. Silberman, D. Hoiem, P. Kohli, R. Fergus. Indoor Segmentation and Support Inference from RGBD Images. *ECCV*, 2012.